Using distance as an Instrumental Variable in LEO to analyse post-16 transitions of disadvantaged young people

Practical workshop

COMMUNITY CATALYST

Youth Transitions

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Thursday 22 October 2026, 10:00–13:00 | 35 Marylebone Road, London

With thanks to the Nuffield Foundation and to FFT Education Datalab

Welcome and housekeeping

- 10:00–13:00 with a 20-minute break at around 11:20
- Practical session: less on findings, more on **how** we accessed, prepared and analysed the data
- Present various tasks needed to complete admin. data study (ONS SRS) using IV; gain feedback on materials most useful to share after event; your own questions
- Slides, selected code and guidance shared on GitHub after the event: we will email you the link
- Please ask questions as we go: this is a workshop, not a lecture.

Agenda

10:00	Welcome and introductions	Peter and all
10:10	1. The study and the data: post-16 pathways and LEO	Peter
10:30	2. From data to code	Min
11:20	Break	
11:35	3. Method: distance as an IV, then bring your own IV	Emma
12:25	4. Results and interpretation, then over to you	Peter
12:45	5. Clinic on your challenges, and support	All
13:00	Close	

Introductions: who is in the room?

INTRODUCTIONS | 10 minutes

- Round the room (about 1 minute each):
  - Your name and organisation
  - The data or research question you are working on (or hope to)
  - **One challenge** you bring today: data access, linking, method or interpretation

- We will come back to your challenges in the ‘over to you’ sessions and the clinic at the end

All

What we hope you take away

- The **step-by-step data and analysis procedures** needed to study post-16 transitions in LEO
- What it is really like to work in the ONS Secure Research Service (SRS), including code you can reuse
- How to implement an **Instrumental Variable (IV)** approach using distance, and the assumptions behind it
- How to interpret IV results for policy, and possible enhancements to the method
- Support available through the ADR England Research Community Catalyst: Youth Transitions

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1

The study and the data

Peter Urwin | 10:10–10:30

Post-16 education pathways of lower attainers

- A key determinant of the post-16 opportunities young people face is their performance in Maths and English GCSEs in the year they turn 16 (KS4)
- Those who do not achieve threshold grades face reduced post-16 choices
- The literature on 'high stakes' exams (e.g. Jerrim, 2022; Machin, McNally and Ruiz-Valenzuela, 2020) concentrates on marginal students
- Our study focuses on young people who miss these thresholds by a **substantial** margin (some work, e.g. Lupton et al., 2021, but a gap in evidence)
- Nuffield Foundation study: Post-16 pathways to employment for lower attaining pupils: are they working? (2024)

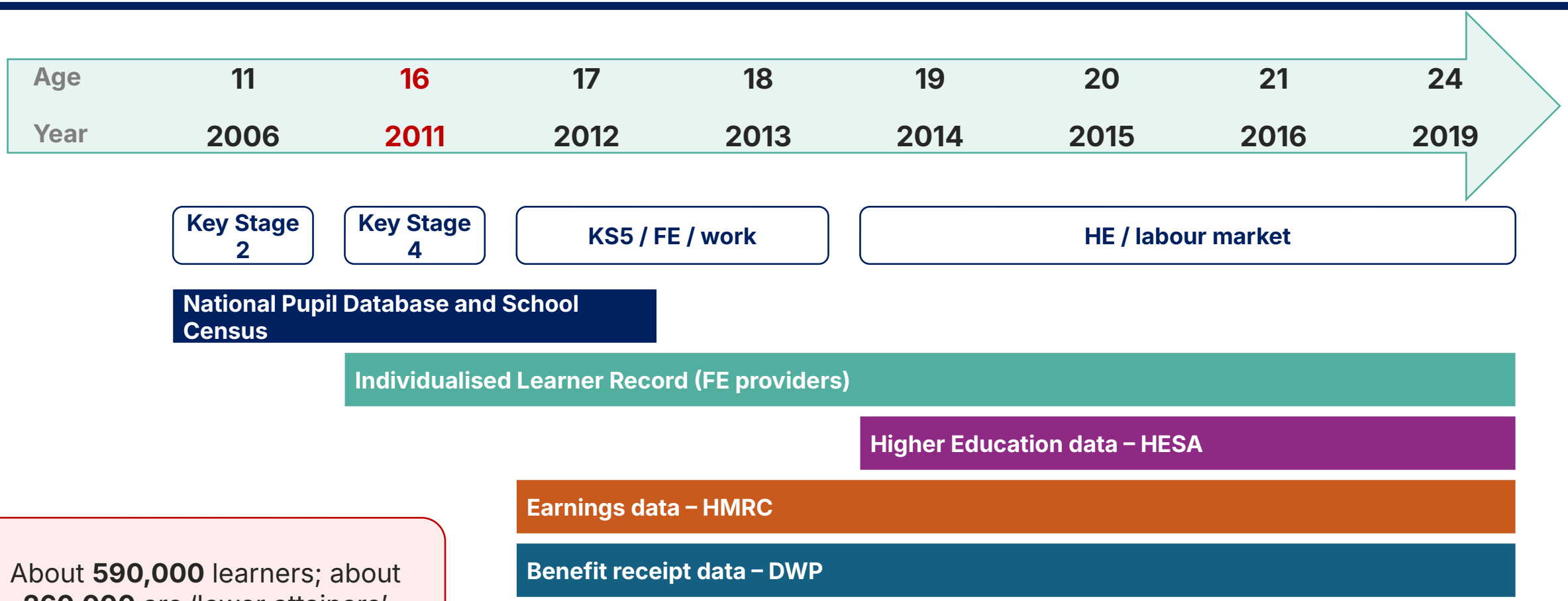
Data and method challenges

- Post-16 policy for young people struggling to reach Level 2 very different from that for those who ‘just miss out’ on GCSE thresholds
- Regression discontinuity methods either side of GCSE thresholds cannot be used
- **Method challenge:** non-random selection into post-16 pathways, potentially related to unobserved factors; no obvious natural experiments
- **Data challenge:** post-16 choices for this group are constrained by institution, and more extensive choice (heterogeneity of qualification treatment) within institutions, when compared to the ‘academic’ route
- Disadvantaged young people at risk of marginalisation face a further ‘missingness’ challenge in the data

PICO: post-16 pathways for lower attainers

P	Population	Lower attaining young people (defined by Maths and/or English GCSE attainment at KS4) in the 2011 education cohort in England, at age 16
I	Intervention	General Further Education (GFE) post-16 pathway (mediation analysis)
C	Comparison	Comparison between pupils in 2011 cohort who take (i) GFE pathway, and those on (ii) alternative post-16 pathways. Using distance as an instrument to emulate random allocation (impacts for 'compliers').
O	Outcomes	•Level 3 or higher (equivalent to two A-level passes) by age 19 •At least one day in employment in the year aged 24 (and variations) •Natural log (Ln) of earnings aged 24

Following the 2011 KS4 cohort through LEO



About **590,000** learners; about **260,000** are 'lower attainers'

National Client Caseload Information System (NCCIS)

Creating post-16 choice sets

- For each lower attaining young person we create a **Choice Set**: includes all post-16 institutions that were a destination for pupils in current and previous 2 cohorts; resident in same MSOA; who also did not achieve Maths and/or English GCSE thresholds.
- For each of these choice-set post-16 institutions [General Further Education (GFE); School or College sixth form; Other vocational] created crow-flies distance.
- Instrument: distance from a young person's place of residence (MSOA population weighted centroid), to nearest post-16 'training provider' in choice set (Dorsett et al., 2019; Urwin et al., 2020; Hendrik Matthewes and Ventura, 2022).

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2

Working in the SRS: from data to code

Min Zhang | 10:30–11:20

Working in the ONS Secure Research Service

- All analysis is carried out within the SRS: **no internet connection**, so no AI-powered tools or web searches for coding help
- Datasets have different access requirements: LEO only via approved office locations or a SafePod (approval can be challenging)
- SafePods must be booked and slots can be congested
- Datasets have different disclosure rules (e.g. LEO and HESA); clearance timeframes vary with ONS staffing.
- For more detail see – Apply to access the Longitudinal Education Outcomes (LEO) dataset [*https://www.gov.uk/guidance/apply-to-access-the-longitudinal-education-outcomes-leo-dataset*]; ONS Data Catalogue [*https://ons.metadata.works/browser/dataset/369/0*]

Working in the ONS SRS: Distance for IV

- External data can only enter via ingestion through pseudonymisation service.
- Crow-flies distances: the main way distance can be used in the SRS, as calculation achieved with application of basic trigonometry
- Geographical distances to colleges need to be created outside the ONS SRS and then ingested via the LEO pseudonymisation service.
- Data ingestion is the secure process by which datasets are brought into the controlled research environment so accredited researchers can analyse them. The ingestion process is tightly regulated under the Digital Economy Act (2017) and the Five Safes Framework.
- Essential that you follow the guidance provided.

LEO in practice: a relational database, not a tidy dataset

- LEO is not an ‘off the shelf’ dataset like a survey: it is a **collection of SQL tables** from different government departments
- You need SQL to link tables using pseudonymised identifiers (e.g. PupilMatchingRefAnonymous in NPD, External_id in ILR, bridged by ae_id)
- Post-16 is the hardest part: school sixth-form pupils are in NPD; FE and sixth-form college learners are in ILR
- Building consistent post-16 pathways across NPD and ILR is why many researchers stop their analysis at age 16
- It does not look like the tidy flow on the previous slide

What LEO is made of *years available in LEO Iteration 2 (variable request form, May 2024)*

Source	Supplied by	Main tables	Years	Person ID
NPD	DfE	School Census, KS2–KS5 attainment, CIN/CLA, absence, exclusions, PLAMS, NCCIS	2001/02–2020/21 (KS2 from 1995/96; NCCIS from 2010/11)	**Pupil reference**
LILR (ILR)	DfE	FE and apprenticeship learners and learning aims	2002/03–2020/21	**ILR learner ID**
HESA	Jisc	Higher education enrolments and outcomes	2004/05–2019/20	**Pupil reference**
HMRC (via DWP)	HMRC	Employment spells, earnings (with IDBR employer), self-assessment, CJRS/SEISS	Employment 1997/98–2020/21; earnings from 2003/04; self-assessment from 2013/14	**EduKey**
DWP	DWP	Benefit spells; home geography; the matching that links everything	Benefits 1999/2000–2020/21	**EduKey**

Activity states

Education and employment

- Education (incl. training): School Census, ILR, HESA; NCCIS education and training codes
- Employment (and self-employment): HMRC; NCCIS employment codes (incl. apprenticeships) and self-employment
- Custody: NCCIS only*

NEET and unknown

- NEET (active): NCCIS codes for young people available for work or learning; our study also used JSA/UC/JTA and the ESA work-related group
- NEET (inactive): NCCIS codes for those not available (e.g. carer, illness); our study also used other benefits with no employment record
- NCCIS other / not known: our study only (the public code drops NCCIS 'not known' codes)
- Not known / missing: no record in any source

From raw LEO tables to analysis: three stages

1. Framework
FFT Education Datalab

- SQL that merges NPD and ILR into consistent post-16 pathways
- Solves the problem of school and FE records that ‘drift apart’
- Shared on GitHub (Youth Transitions Resource Hub)
- Data Explained: Understanding post-16 learning spells and qualification aims to estimate returns to learning

2. Construction
Our data manipulation

- Education variables built for the research question
- E.g. classroom v non-classroom, vocational v academic, indicators by age 19 and 21
- Treatment rules: how many months in FE v school sixth form; which qualification combinations

3. Analysis
Estimation

- OLS and IV (2SLS) models
- First stage and diagnostics
- Interpretation (Sections 3 and 4)

How the Datalab scripts fit together

Source loaders: one script per dataset

- **School Census** — census_details
- **Exclusions** — exclusions
- **Absence** — attendance
- **CIN / CLA spells** — in_need, looked_after
- **School sixth-form aims** — plams
- **FE learning aims** — ilr_aims
- **Higher education** — hesa
- **NCCIS activity** — nccis_spells
- **Employment** — employment
- **Earnings** — earnings
- **Benefits** — benefits

Derived tables

- **census** — School entry and leaving dates
- **pathways** — Programme of study
- **metrics** — Days in each activity

What they give you

- Pupil characteristics, pupil cohort, dates in school
- Pathway type on October, January and May census days
- Days in school, FE, HE, work, benefits and NCCIS states; annual earnings

person_lookup
Every loader adds its people here: ae_id links pupil records (NPD, HESA), ILR learners and HMRC/DWP records into one fft_person_id

SQL Server scripts run inside the SRS. Written for LEO iteration 2; source: github.com/Youth-Transitions/resources

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What each source script loads

Script	What it does	One row per	Person ID in the source
census_details	Every School Census term, plus AP and PRU censuses	Pupil, term and school	**Pupil reference (NPD)**
attendance	Absence: sessions possible and missed	Pupil, term and school	**Pupil reference (NPD)**
exclusions	Fixed-term and permanent exclusions	Exclusion	**Pupil reference (NPD)**
in_need_spells, looked_after_spells	Periods as a child in need or looked after	Spell	**Pupil reference (NPD)**
plams	Post-16 qualifications studied in school sixth forms	Qualification, year, school	**Pupil reference (NPD)**
nccis_spells	Local authority monthly activity records, as spells (incl. NEET)	Activity spell	**Pupil reference (NPD)**
hesa	Higher education enrolments	HE record and year	**Pupil reference (NPD)**
ilr_aims	FE and apprenticeship learning aims	Learning aim and year	**ILR learner ID**
employment	Employment spells (jobs with the same dates combined)	Employment spell	**EduKey (HMRC/DWP)**
earnings	Annual earnings from PAYE and self-assessment	Person and tax year	**EduKey (HMRC/DWP)**
benefits	Benefit spells, flagged out-of-work or other	Benefit spell	**EduKey (HMRC/DWP)**

How records are linked across datasets

1 Matching by government

DfE matches education records to DWP's records of everyone with a National Insurance number, using name, date of birth, sex and postcode (plus NI number where held, e.g. ILR).

Researchers never see these details and cannot re-match.

2 Each source keeps its own ID

Pupil reference
NPD school records, NCCIS, HESA

ILR learner ID
Further education (ILR)

EduKey
HMRC employment and earnings; DWP benefits

3 ae_id

LEO's bridging tables link each source ID to one person

4 fft_person_id

Datalab's person_lookup gives one number per person, used by every table

Derived tables are keyed by fft_person_id

pathways	metrics_wide	metrics_monthly	census_dates (optional)
Person and census date	Person and year	Person and month	Person, term and school

Code walk-through 1: tidy each source

fft.employment.sql (lines 202–220)

```
insert fft.employment
select
    max(leo_edukey),
    startdate,
    enddate,
    fuzzy_start_date = min(uncertainstartdate+0),
    fuzzy_end_date = min(uncertainenddate+0),
    fft_person_id
from
    LEO_i2."<project number>".leo_employment e                    1
    left join fft.person_lookup k                                 2
        on    e.edukey collate Latin1_General_BIN = k.leo_edukey
where
    fft_person_id is not null
group by
    fft_person_id,                                                3
    startdate,
    enddate
;
```

‘Sam' appears in three unconnected records

Invented example

School Census · Pupil ref P-0001
On roll at School A, autumn 2011

ILR (college) · Learner ID L-5521
Level 2 course at College B, from Jan 2012

HMRC employment · EduKey E-7734
Job from Oct 2012

What this script does

1. **Read the HMRC employment records**
2. **Attach each person's number**
3. **Same person, same dates: one spell**

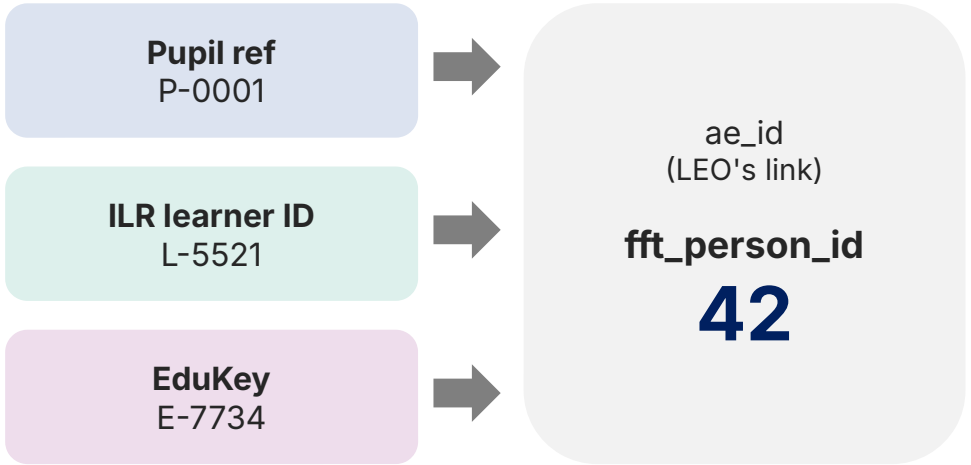
Code walk-through 2: join the records into one person

fft.ilr_aims.sql (lines 140–147): the person lookup

```
create table fft.person_lookup (
    fft_person_id int not null,          -- 2
    multiplex tinyint not null,
    ae_id bigint,                        -- 1
    pupil_matching_reference char(18) collate Latin1_General_BIN,
    ilr_external_id bigint,
    leo_edukey char(14) collate Latin1_General_BIN
    );
```

In plain English
One row per person. Each column holds that person's ID in a different dataset, so any record can be traced to the same person number.

Invented example



1 ae_id comes from the government's matching; researchers cannot change it
2 Datalab give each ae_id one simple number, used in every other table

Code walk-through 3: build the timeline

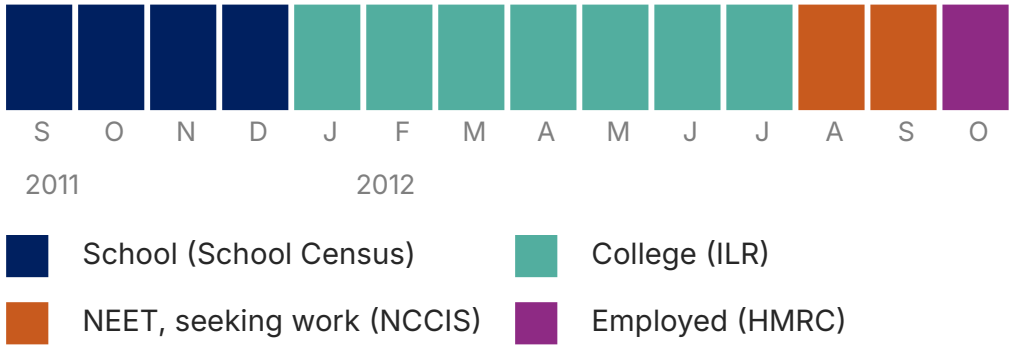
fft.metrics.sql (lines 134–152)

```
create table fft.metrics_monthly (
    fft_person_id int not null,
    start_date date not null,                                    1
    end_date as dateadd(d, -1, dateadd(m, 1, start_date)),
    days_in_school tinyint,                                      2
    days_in_further_education tinyint,
    days_in_higher_education tinyint,
    days_in_employment tinyint,
    days_on_in_work_benefits tinyint,
    days_on_out_of_work_benefits tinyint,
    days_in_custody tinyint,
    days_in_education_nccis tinyint,
    days_in_training_nncis tinyint,
    days_in_employment_nccis tinyint,
    days_self_employed_nccis tinyint,
    days_active_neet_nccis tinyint,                              3
    days_inactive_neet_nccis tinyint,

    primary key (fft_person_id, start_date)
```

Sam's months after Year 11

Invented example



1. **One row per person per month**
2. **Days in each source: school, FE, HE, work, benefits**
3. **NCCIS kept as separate columns, incl. NEET**

This is the table researchers then analyse.

When sources disagree

Where conflicts come from

School roll	vs	**FE learning aim**
Same young person, same census date		
NCCIS: NEET	vs	**HMRC: employed**
Local authority record vs tax record		
DWP: out-of-work benefit	vs	**HMRC: employed**
Overlapping spells in the same month		
NCCIS: one activity	vs	**NCCIS: another activity**
Two local authorities, same start date		

DfE LEO pathways (2021)

One main activity per tax year, based on sustained spells (e.g. 6 consecutive months)

Priority when activities overlap

Education → **Out-of-work benefits** → **Employment**

Within education: KS5 > HE > adult FE
Leftovers: 'no sustained activity' and 'activity not captured'

The hierarchy is an analytical choice: state it, and check whether results change under alternatives.

Source: DfE (2021) Technical report for education and labour market pathways of individuals (LEO)

How Datalab handles conflicts

FFT Education Datalab (2026)

Initial destination at 16 on one census date (6 October 2022)

1. **On a school roll or an FE course? Use School Census and ILR**
2. **Otherwise, use the NCCIS activity on that date**
3. **If 'not known', check later NCCIS records that started earlier**

Result: 1.5% to 3.6% of 16-year-olds may be NEET, depending on assumptions

Inside NCCIS

1. **Custody overrides everything else**
2. **Two different activities with the same start date: the record is dropped**
3. **'Not known' records are dropped**
4. **Gaps between activities are closed at a random point**

Across sources
Datalab's public code keeps each source side by side; it does not choose one main activity.

Sources: FFT Education Datalab (2026) Using administrative data to study initial post-16 destinations; Youth Transitions resources (NCCIS script and wiki)

What should we prioritise sharing?

OVER TO YOU | 10–15 minutes | whole group or tables

- Having seen the framework and an example, which parts would be most useful for us to prioritise sharing on GitHub?
  - Coding education variables and qualification categories
  - Cross-institution indicators and post-16 pathways
  - Activity states (including NEET and ‘not known’) and activity measures
  - Choice sets, distance and the IV estimation code
- Tell us your specific questions

All

Which LEO? Data currency and code caveats

- Current SRS extract: LEO Iteration 2.1 (distributed May 2026): education data to 2020/21; latest tax and benefit data to March 2024.
- DfE suggest they are working towards iteration 3 in Summer 2027.
- Therefore, currently there is no post-Covid NPD data in LEO (though there is post-Covid NPD data in SRS - just not linked to LEO).
- Our code was written for an earlier iteration of LEO, so expect table and variable differences.
- Stay in touch with LEO developments, e.g. the UCL LEO user group (20th October 2026 latest meeting)
- Also see shared LEO SQL code library through SRS.

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Break

11:20–11:35

3

Method: distance as an Instrumental Variable

Emma Gorman | 11:35–12:25

Back from the break: our question and our data

Research question

For young people who leave school with low Maths and English GCSE grades, does going to a **general FE college (GFE)** at 16 improve their qualifications, employment and earnings?

Treatment (D)

Registered at a GFE in October of the year they turn 17

Comparison: everything else (sixth form, other learning, work, NEET)

Outcomes (Y)

Level 3 by age 19

In employment at age 24

Log earnings at age 24

Controls (X)

KS2 and KS3 attainment

FSM, SEN, ethnicity, gender

Region, urban/rural, MSOA % FSM and income decile

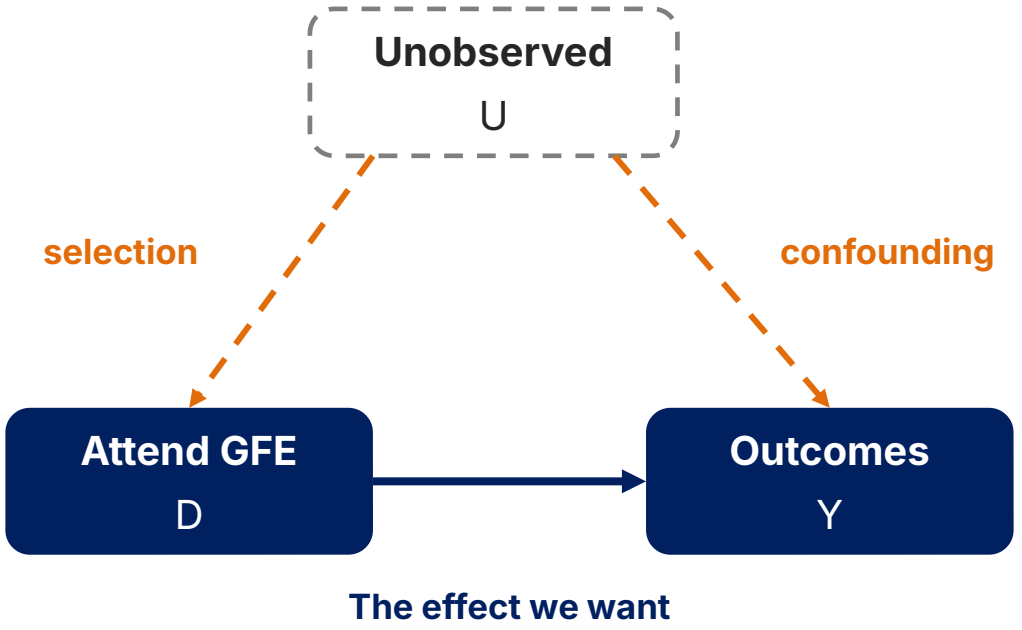
Instrument (Z)

Distance from home (Year 11 MSOA) to the nearest GFE in the young person's choice set

Population: 2011 KS4 cohort with E, F or G in both Maths and English (about 47,000), followed in LEO to age 24

The identification problem in a nutshell

- Young people are not randomly allocated to GFE: they choose, and are chosen
- Things we cannot see in LEO, such as motivation, family support, health or local job opportunities, may drive **both** the choice of GFE **and** later outcomes
- So comparing GFE with non-GFE learners, even with rich controls, is likely to be biased
- The bias could go either way



We need something that moves people into GFE for reasons unrelated to their outcomes

An instrument: an imperfect randomised trial

- In an RCT, a coin toss decides who is treated, and (ideally) everyone complies
- An instrument is a 'fuzzy' coin toss: something **as good as random** nudges people towards treatment
- Not everyone responds to the nudge: some always take the treatment, some never do
- IV learns only from the people the nudge switches: the **compliers**

Heterogeneous treatment effects framework

- Angrist, J. D., & Pischke, J.-S. (2009). *Mostly harmless econometrics: An empiricist's companion*. Princeton University Press
- Cunningham, S. (2021). *Causal Inference: The Mixtape*. Yale University Press. https://doi.org/10.12987/9780300255881

Example: distance to GFE

Always-takers	Go to GFE whether it is near or far
Compliers	Go to GFE if it is near, but not if it is far
Never-takers	Do not go to GFE, wherever it is

IV estimate = effect of GFE for compliers

Instrumental variables - examples

Example	Instrument (Z)	Treatment (D)	Outcome (Y)
Vietnam draft lottery (Angrist, 1990)	Low lottery number, drawn at random by birthday	Military service	Later earnings
Flu jab reminders (McDonald et al., 1992)	Patient's doctor randomly sent reminders to vaccinate	Flu vaccination	Flu-related hospital admission
Raising the school leaving age (Gorman, 2023)	Born on or after 1 September 1957	An extra year of school	Memory in later life
Our study	**Distance to a GFE college**	Attending GFE	Level 3, employment and earnings

Two-stage least squares (2SLS)

Stage 1 (first stage): predict GFE from the instrument and the controls

$$\text{GFE}_i = \pi_0 + \pi_1 Z_i + \gamma_1 x_{1i} + \gamma_2 x_{2i} + \ \ldots \ + \nu_i$$

Stage 2: replace GFE with its fitted values from stage 1

$$\text{Employment}_i = \beta_0 + \beta_1 \widehat{\text{GFE}}_i + \delta_1 x_{1i} + \delta_2 x_{2i} + \ \ldots \ + \varepsilon_i$$

β_1 is the effect we want: it uses only the variation in GFE that is driven by distance

The same controls (x) go in both stages; Z appears only in the first

In practice, run both stages in one command (ivreg2, ivregress, feols): doing it by hand gives the wrong standard errors

Five IV assumptions

1. First stage (relevance)

The instrument really does shift who gets the treatment (non-zero correlation and robust F-stat from first-stage)

2. Independence

The instrument is 'as good as' randomly assigned (given our controls)

3. Exclusion restriction

The instrument affects the outcome **only** through the treatment

4. Monotonicity

The instrument works in one direction for everyone: no 'defiers'

5. SUTVA

No spillovers: my outcome depends on my own instrument and treatment, not on yours

Put together

IV estimates the average effect for compliers: the Local Average Treatment Effect (LATE)

Distance as an instrument: the literature

- A good IV is hard to find ‘in the wild’: it needs a careful argument based on theory and empirics
- Instrument: distance from a young person’s home to the nearest relevant post-16 provider (Dorsett et al., 2019; Urwin et al., 2020; Hendrik Matthewes and Ventura, 2022)
- Some studies challenge distance to college as an instrument: in some contexts it correlates with ability and family background (Kjellström and Regner, 1999; Cameron and Taber, 2004)
- IV estimates identify Local Average Treatment Effects (LATE) for ‘compliers’

Applying the assumptions to distance (1)

First stage (relevance)

- Distance should predict GFE attendance: living further away lowers the chance of attending
- Report the first-stage coefficient and its robust F-statistic (available in IV statistical packages)

Independence

- Local economic conditions could be correlated with FE, distance and earnings
- We control for detailed urban/rural classification and MSOA characteristics (e.g. % FSM)
- Self-selection: do families likely to benefit from FE live closer?
- We use distance to the nearest FE that will accept you, given your GCSEs (the choice set) – hard to predict in advance

Applying the assumptions to distance (2)

Exclusion restriction

- Distance affects the probability of attending GFE
- We argue it is unlikely to affect employment or earnings directly

Monotonicity and SUTVA

- Living closer to FE should only increase, never decrease, the chance of attending
- SUTVA: could college capacity or peer effects mean one young person's attendance affects another's outcomes?

Practical steps: creating post-16 choice sets

Each young person gets a list of realistic post-16 options. The instrument is the distance to the nearest General FE college on that list.

1. **Who**
   Young people who missed grade C (4) in Maths and/or English. Choice sets are built only for them

2. **Look back**
   Similar young people from the same MSOA, in the current and previous two cohorts

3. **The choice set**
   Every institution where any of them was in October of the year they turned 17

4. **The instrument**
   Distance from home (MSOA) to the nearest General FE college in the set

Treatment: attending a General FE college in October of the year they turn 17. Young people not at any institution in the set stay in the comparison group.

Choice set for young people in MSOA X

Invented example

Institution	Type	km	Went there?*
College A	General FE	6.1	Yes: in set
School B	School sixth form	1.1	Yes: in set
College C	Sixth form college	5.4	Yes: in set
College D	**General FE**	**3.2**	**Yes: in set**
College E	General FE	2.0	No: not in set

Instrument: 3.2 km (College D, the nearest General FE college in the set). School B is closer but is a school sixth form. College E is closer still, but no similar young people from MSOA X went there, so it is not in the set.
*Young people from MSOA X who also missed the thresholds, current and previous two cohorts, in October at 17.

Practical steps: crow-flies distance

The instrument is the straight-line ('crow-flies') distance from home to the nearest General FE college in the choice set.

1. **Locations**
   Each institution, and the centre of the young person's MSOA, as eastings and northings (metres on the national grid)

2. **Calculate**
   Straight-line distance by simple trigonometry: Britain is small enough for a flat map grid

3. **Bring into the SRS**
   Calculated outside the SRS and ingested via the LEO pseudonymisation service

4. **Link**
   Each young person gets the distance to the nearest General FE college in their choice set

We are not sharing distance code at this event.

Worked example: MSOA X to College D

Point	Easting (m)	Northing (m)
MSOA X centre	530,000	180,000
College D	532,000	182,500
Gap	2,000	2,500

$$\sqrt{(2{,}000^2 + 2{,}500^2)} = 3{,}202 \text{ m} \approx 3.2 \text{ km}$$

Invented example

In the analysis

- In km; kept within 25 km of the nearest General FE college
- Not a straight-line relationship with attending: log and bands also tried
- Run separately for rural and urban areas

Min Zhang

Back to the IV code: now we can read it

Each part of Min's code maps onto an idea from this section

Simplified from 5.IV_prototype_2011_cohort_v5_subgroup_loop.do (Stata)

```
forvalues i = 0/9 {
    if `i'==0 local cond "1"                    // everyone
    if `i'==1 local cond "hard_fail == 1"       // E, F or G in both
    if `i'==4 local cond "soft_fail == 1"       // two Ds
    if `i'==8 local cond "at_least_one_D == 1"  // at least one D

    foreach v of varlist $eduout {
        reg    `v' attfe $controls1 if `cond', robust
        est sto ols_`v'
        ivreg2 `v' (attfe = dist_km) $controls1 if `cond', robust savefirst
        est sto iv_`v'
        putexcel B`p' = `e(widstat)'
        ivreg2 `v' (attfe = dist_km) $controls1 if `cond' & urban2==0, robust savefirst
        est sto ivR_`v'
    }
    esttab ols_* using "${outdir}/ols.csv", $spec3
    esttab iv_*  using "${outdir}/basic_ivs.csv", $spec3
    esttab ivR_* using "${outdir}/iv_rural.csv", $spec3
    estimates clear
}
```

1. **Loop over subgroups**
   Each subgroup has its own compliers, so its own LATE
2. **Loop over outcomes**
   Same instrument; first stage only changes if the sample does (e.g. earnings)
3. **OLS v 2SLS**
   reg uses all variation in GFE; ivreg2 uses only the part driven by distance
4. **First stage**
   e(widstat): is distance a strong enough nudge?
5. **Rural and urban**
   Does distance mean the same thing everywhere?
6. **Store and tabulate**
   One set of tables per subgroup, then clear

Bring your own IV

OVER TO YOU | 10–15 minutes | whole group or tables

- Do you have a treatment, an outcome and a possible instrument in your own work?
- Which of the five assumptions is most at risk, and what evidence would help?

All

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4

Results and interpretation

Peter Urwin | 12:25–12:45

Results: OLS and IV

Treatment: registered with a GFE provider in October of the academic year turning 17
Population: 2011 KS4 pupils achieving E, F or G grades in Maths and English GCSE

Outcome	Outcome mean	OLS	IV (2SLS)
Level 3 (NQF) by age 19	0.07	**0.023***** (0.002)	**0.097*** (0.048)
At least one day in employment in the year aged 24	0.66	**0.045***** (0.005)	**-0.177*** (0.089)

Standard errors in brackets. *** p<0.001, * p<0.05. Source: Bibby et al. (2024), Nuffield Foundation.

Interpreting IV estimates

- IV estimates identify a Local Average Treatment Effect (LATE) that applies to the subpopulation of compliers (Imbens and Angrist, 1994) — i.e. young people whose engagement with post-16 learning is altered by distance to provider.
- Only a portion of the variation in the endogenous variable ‘GFE’ (portion matching up with the ‘distance’ instrument) is used to produce slope estimate.
- Think of this as capturing the treatment impact associated with the ‘marginal’ learner (i.e. they are on the margins of going to GFE, or not).
- This provides us with a Local Average Treatment Effect [LATE] as opposed to the Average Treatment Effect (ATE/ATT) from OLS.

Who is the counterfactual? Interpreting OLS and IV together

- OLS covers the whole population, but selection on unobservables may drive the positive estimates
- IV should capture the causal effect for compliers, but the alternative they would otherwise take varies widely
- Compliers did not go to GFE partly because it was further away: their alternative could be work (formal or informal), other learning, or being out of education and work
- Other groups (outside compliers) may still benefit, and this could drive the OLS results

Different counterfactuals: margin-specific effects

- Our LATE compares GFE with an amalgam of alternatives: school sixth form, other learning, work, NEET
- With more than one instrument (e.g. distance to GFE **and** distance to a school sixth form) we could separate margins
  - GFE v no learning: young people who would otherwise not have continued in education
  - GFE v sixth form: young people diverted from another education route
- Example: Mountjoy (2022) separates the effects of community colleges for students diverted from four-year colleges and for those who would otherwise not have attended college
- Requires stronger assumptions and plenty of variation in both distances

Conclusions and possible enhancements

- Results suggest that, at best, there is no significant employment return from post-16 learning for many of the lowest attaining young people, relative to a ‘no learning’ alternative
- Much has changed in post-16 policy since 2011, but mediation analysis suggests this has not improved outcomes for lower attainers
- **Enhancement:** Marginal Treatment Effects, to examine how LATE estimates vary across the population that underpins the OLS results (Carneiro, Heckman and Vytlacil, 2011; Dorsett and Stokes, 2021)
- **Enhancement:** margin-specific treatment effects, comparing GFE with each alternative (previous slide)

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5

Clinic and support

All | 12:45–13:00

Clinic: your challenges

- Back to the challenges you raised at the start
- Data access and the SRS (Min)
- Method and assumptions (Emma)
- Interpretation and policy (Peter)
- Anything we cannot answer today: we will follow up after the event

Discussion

Share your experience

All

Youth Transitions Resource Hub

- https://github.com/Youth-Transitions/resources/wiki
- A guide and example scripts to set up LEO to study post-16 activities
- Calculates the activity measures used in this presentation
- After the event: slides, selected code and guidance from this session

Support in data and analysis

- **Data Explained guides**, e.g. LEO: understanding post-16 learning spells and qualification aims; identifying and studying young people who are NEET
- More on Geographic distance, FSM/IMD, ONS-LS.
- Understanding appropriate administrative datasets (prototype tool)
- FFT Education Datalab training (e.g. introductory guidance using SQL) and the UCL training programme
- Online clinics for researchers new to the data, and one-to-one sessions
- Forthcoming Westminster sessions on quasi-experimental approaches using administrative data

How to get involved

Research conferences	Online forum	Newsletters	Resource Hub
• Focus on policy, practice and data landscape • Engagement with the research agenda • Learning, development, collaboration	• Active communication channel • Announcements • Semi-formal setting for technical questions	• Latest research from across the community • Peer review • Upcoming events and activities	• One-stop shop for all resources • New content throughout the project

adruk.org: learning hub, user groups and communities, ADR England Research Community Catalyst: Youth Transitions

A

Appendix: LEO datasets, activity states and missingness

Reference slides

LEO datasets for studying post-16 activities

Education	Labour market	NEET
School Census	Employment	
ILR	Benefits	
HESA	Earnings and self-employment	

NCCIS (local authority tracking of 16–17-year-olds): education, employment, NEET and ‘not known’ status

- Each dataset contributes something, but each has its own quirks and they can conflict
- All observe days, apart from earnings (annual); NEET is often a residual category

Activity states (using NCCIS)

Education and employment

- Education (incl. training): School Census, ILR, HESA
- Employment (and self-employment)
- Custody

NEET and unknown

- NEET (active): out-of-work benefits (JSA/UC/JTA), ESA work-related group, NCCIS codes
- NEET (inactive): other out-of-work benefits, other benefits with no employment record, NCCIS codes
- NCCIS other / not known
- Not known / missing

Activity measures (reference)

- Number of days per year (academic or tax) in each activity
- Number of days per month in each activity, and a ‘main’ activity each month
- Activity flags on enumeration dates: 31 October each year and termly School Census dates
- These measures feed both the descriptive pathways and the definition of treatment

A first look at the IV code

For each subgroup, run every outcome, then save the tables. Emma explains the method after the break.

Simplified from 5.IV_prototype_2011_cohort_v5_subgroup_loop.do (Stata)

```
forvalues i = 0/9 {
    if `i'==0 local cond "1"                    // everyone
    if `i'==1 local cond "hard_fail == 1"       // E, F or G in both
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    foreach v of varlist $eduout {
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        ivreg2 `v' (attfe = dist_km) $controls1 if `cond', robust savefirst
        est sto iv_`v'
        putexcel B`p' = `e(widstat)'
        ivreg2 `v' (attfe = dist_km) $controls1 if `cond' & urban2==0, robust savefirst
        est sto ivR_`v'
    }
    esttab ols_* using "${outdir}/ols.csv", $spec3
    esttab iv_*  using "${outdir}/basic_ivs.csv", $spec3
    esttab ivR_* using "${outdir}/iv_rural.csv", $spec3
    estimates clear
}
```

1. **Loop over subgroups**
   Outer loop: ten subgroups, e.g. at least one D
2. **Loop over outcomes**
   Inner loop: each outcome, e.g. Level 3 by 19
3. **OLS and 2SLS**
   Distance to the nearest GFE instruments attending GFE
4. **Instrument strength**
   First-stage F statistic saved for each model
5. **Subsamples**
   The same model for rural and urban areas
6. **Store and tabulate**
   One set of tables per subgroup, then clear

Min Zhang

Activity states (using NCCIS)

Education and employment

- Education (incl. training): School Census, ILR, HESA; NCCIS education and training codes
- Employment (and self-employment): HMRC; NCCIS employment codes (incl. apprenticeships) and self-employment
- Custody: NCCIS only

NEET and unknown

- NEET (active): NCCIS codes for young people available for work or learning; our study also used JSA/UC/JTA and the ESA work-related group
- NEET (inactive): NCCIS codes for those not available (e.g. carer, illness); our study also used other benefits with no employment record
- NCCIS other / not known: our study only (the public code drops NCCIS 'not known' codes)
- Not known / missing: no record in any source

The ‘not known’ group

Genuinely NEET or not observed

- NEET (inactive)
- NEET (active, but not claiming benefits)
- (E)migration, or death

Matching issues

- Under-matching between NPD and HMRC/DWP
- Over- or under-matching between NPD, ILR and HESA, and between DWP and HMRC
- Earnings records without employment records
- About 95% of education records match to DWP/HMRC (98% non-EAL; 88% EAL)

Data strengths and limitations

- Benefits data capture NEET on ‘inactive’ or ‘active’ benefits
- Depending on cohort, age and group, 20% or more may be ‘missing’
- NCCIS improved identification immediately post-16, but with regional variability due to administrative practice
- From age 19 there is no NCCIS to separate e.g. economically inactive from living abroad or deceased
- Studying ‘pathways out of NEET’; strategies to impute status; recent cohorts have less time post-NCCIS

NCCIS activity codes

Education
210 - Full-time education - school sixth-form
220 - Full-time education - sixth-form college
230 - Full-time education - further education
240 - Full-time education - higher education
250 - Part-time education
260 - Gap year students
270 - Full-time education - other
280 - Special post-16 institution
290 - Full-time education - custodial institution (juvenile offender)

Training
410 - ESFA funded work-based learning
430 - Other training
440 - DWP training and support programme
450 - Traineeship
460 - Supported Internship

Re-engagement activities
530 - Re-engagement provision

Employment
310 - Apprenticeship
320 - Full-time employment with study (regulated qualification)
330 - Employment without training
340 - Employment with training (other)
350 - Temporary employment
360 - Part-time employment
380 - Self-employment
381 - Self-employment with study (regulated qualification)
550 - Work not for reward with study (regulated qualification)

NEET
540 - Working not for reward
610 - Not yet ready for work or learning
615 - Start date agreed (other)
616 - Start date agreed (RPA compliant)
619 - Seeking employment, education or training
620 - Not available to labour market/learning - carer
630 - Not available to labour market/learning - teenage parent
640 - Not available to labour market/learning - illness
650 - Not available to labour market/learning - pregnancy
660 - Not available to labour market/learning - religious grounds
670 - Not available to labour market/learning - unlikely ever to be economically active
680 - Not available to labour market/learning - other reason

Other
710 - Custody (young adult offender)

Current situation not known
810 - Current situation not known
820 - Cannot be contacted - no current address
830 - Refused to disclose activity